APPLIED ML/AI · FINAL YEAR PROJECT PROPOSAL

Predicting and Simulating Schedule and Cost Risk in NYC Capital Projects

An early-warning classifier and a data-driven Monte Carlo simulation for public infrastructure portfolios

Student	Shubh Desai	**Course**	Applied ML/AI
Student ID	0990724	**Instructor**	Prof. Weijia Peng
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Abstract

Public infrastructure projects are chronically delivered late and over budget, yet the planned completion dates and budgets that agencies publish rarely reflect that history. This project builds a two-stage, data-driven approach to the problem using the NYC Open Data *Capital Projects* dataset (3,320 reporting snapshots of 363 projects, 2013–2023). In the first stage, supervised classifiers (logistic regression, Random Forest and XGBoost) predict whether a project will slip or overrun, using **only information available at each reporting date**, and SHAP values explain the drivers. In the second stage, a Monte Carlo simulation replaces guessed duration ranges with **empirical phase-level slip rates**, resampled by agency and phase, to estimate each high-risk project's probability of on-time delivery, its P50/P80 completion dates and the schedule and budget contingency it needs. A data audit found target leakage and exposure bias in the raw data and motivated a fixed-horizon target. Treating every report as a prediction point (998 examples) and modelling delay and overrun separately raises grouped cross-validated ROC-AUC from 0.67 to **0.73 (delay)** and **0.76 (overrun)**. A back-test on 340 historical cases shows the simulator's P80 holds 78% of the time (target 80%), with a median error **42% lower** than trusting the reported completion date. The final deliverables are a reproducible Python pipeline, a decision dashboard, and validated contingency guidelines.

Keywords: project risk analytics; cost overrun; schedule delay; early warning; explainable machine learning; SHAP; Monte Carlo simulation; reference class forecasting; public infrastructure

1. Introduction and Motivation

Cost and schedule overruns are among the most persistent problems in public works. In a study of 258 transport infrastructure projects, Flyvbjerg, Holm and Buhl (2002) found that nearly nine in ten exceeded their budgets, with an average overrun of 28%, and that the pattern had not improved over 70 years. Flyvbjerg (2014) later summarized this as the "iron law" of large projects: *over budget, over time, over and over again*. The explanation is partly behavioral. Planners anchor on the specifics of their own project and underweight the outcomes of similar past projects, the *planning fallacy* described by Lovallo and Kahneman (2003).

Two remedies follow from this literature. The first is **early identification** of the projects most likely to go wrong, so that review effort is focused where it matters. The second is **reference class forecasting** (Flyvbjerg, 2006): replacing a project's own optimistic estimate with the empirical distribution of outcomes from comparable projects. Industry practice already uses Monte Carlo simulation for integrated cost and schedule risk analysis (AACE International, 2011), but the input ranges are usually elicited from expert judgment rather than measured.

New York City publishes periodic snapshots of its capital project portfolio, recording each project's phase, budget forecast and forecast completion date at every reporting period. That history makes it possible to apply machine learning to the first remedy and to **derive simulation inputs directly from data** for the second. This project combines both into a single pipeline.

2. Problem Statement, Research Questions and Hypotheses

Problem. Can a city identify high-risk capital projects early, and keep re-assessing them as they progress, and quantify how much schedule and budget contingency they realistically need?

Table 1. Research questions and testable hypotheses.

	Research question	Hypothesis
RQ1	Which project factors (managing agency, category, size, phase, planned duration) are associated with delays and cost overruns?	**H1:** Projects that enter the portfolio while still in design carry higher risk than those entering at procurement or construction.
RQ2	Using only information available at a given report, can a model rank projects by their risk of delay and of overrun over the next two years?	**H2:** Grouped-CV ROC-AUC is significantly above 0.5, and re-scoring at every report improves on scoring at entry only.
RQ3	For a high-risk project, what is the probability of meeting its reported completion date, and what contingency is needed for 80% confidence?	**H3:** Reported dates of high-risk projects have under a 50% chance of being met, and a data-driven simulator is better calibrated than the reported dates.

3. Objectives and Expected Contributions

- **A leakage-free early-warning model.** Separate delay and overrun classifiers that re-score every project at every report, validated with grouped cross-validation and a temporal holdout, and explained with SHAP.

- **Data-driven simulation inputs.** Phase-level slip and cost-growth rates measured from report-to-report changes, so the Monte Carlo model is a quantitative form of reference class forecasting.
- **A validated simulator.** A back-test of the Monte Carlo model against real historical outcomes, with an out-of-sample recalibration step, so every contingency figure carries a measured error rate.
- **Decision outputs.** P50/P80 completion dates, schedule and budget contingency, tornado sensitivity analysis, and a portfolio table of recommended buffers by agency and phase.
- **A documented data audit.** Identification of leakage and exposure bias in the public dataset, with corrective target design, which is a methodological contribution in its own right.
- **Reproducible artefacts.** Jupyter notebooks, clean data tables and an interactive Power BI/Tableau dashboard.

4. Data

The study uses the NYC Open Data *Capital Projects* dataset (file `Capital_Projects_20261025.csv`). The grain is one row per project per reporting snapshot: 3,320 rows, 363 projects and 24 snapshots between September 2013 and January 2023, across 11 managing agencies (DDC, DOT, DEP, EDC and SCA are the largest).

Table 2. Dataset fields and their role in the analysis.

Field group	Fields	Use in this project
Identity	PID, project name, description	Join key and labelling
Context	Managing agency, client agency, category, borough	Categorical features; dashboard slicers
Status	Current phase, design start	Phase features; phase-level slip attribution
Budget	Budget forecast; latest and total budget changes	Cost targets; cost-growth rates
Schedule	Forecast completion; latest and total schedule changes	Delay targets; schedule slip rates

4.1 Preliminary data audit

An initial audit (complete) surfaced three issues that directly shape the methodology:

- **Leakage in the "total change" fields.** *Total Schedule Changes* and *Total Budget Changes* never vary within a project: every snapshot repeats the lifetime total, including snapshots taken before the changes occurred. They equal the sum of the per-report changes in 87% of projects and correlate 0.97 with the observed movement of the forecast date. Using them as features would leak the outcome; the true history lies in *Forecast Completion* and *Budget Forecast*, which change between reports.

- **Exposure bias.** Projects entered the dataset at different times (about 20 projects in 2013, 363 by 2023). Lifetime delay therefore measures how long a project was observed as much as how risky it was (Spearman $\rho = 0.77$ between years tracked and delay; Figure 2a).
- **Reporting gaps.** SCA reports zero schedule change in 63% of rows and FDNY in 100%, versus 2% or less for other agencies; borough is 55% missing. SCA and FDNY are excluded from the delay model.

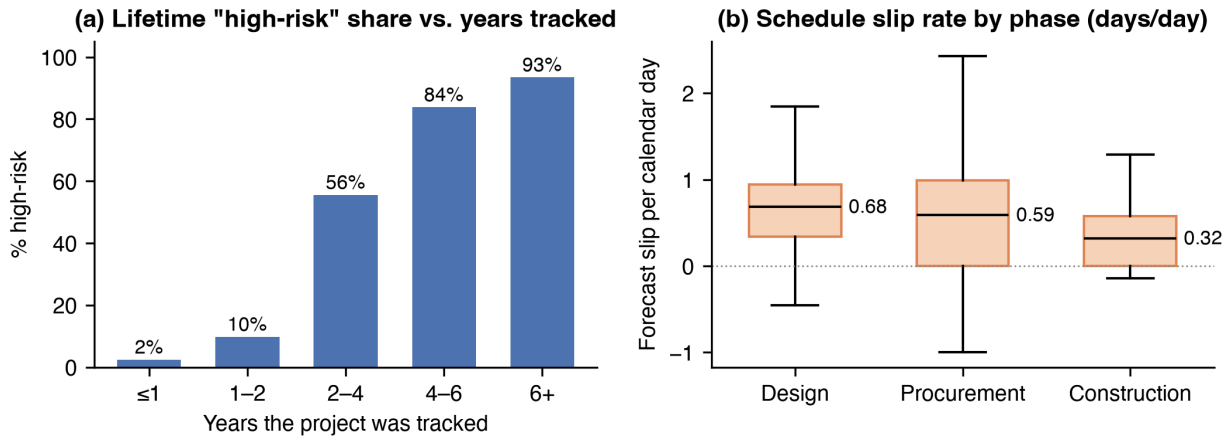


Figure 1. (a) The share of projects that look "high-risk" over their lifetime rises from 2% to 93% with the number of years tracked, which motivates a fixed observation window. (b) Measured schedule slip rates by phase (days the forecast completion moved per calendar day in the phase; 439 project–phase records). Design slips fastest (median 0.68), construction slowest (0.32).

5. Methodology

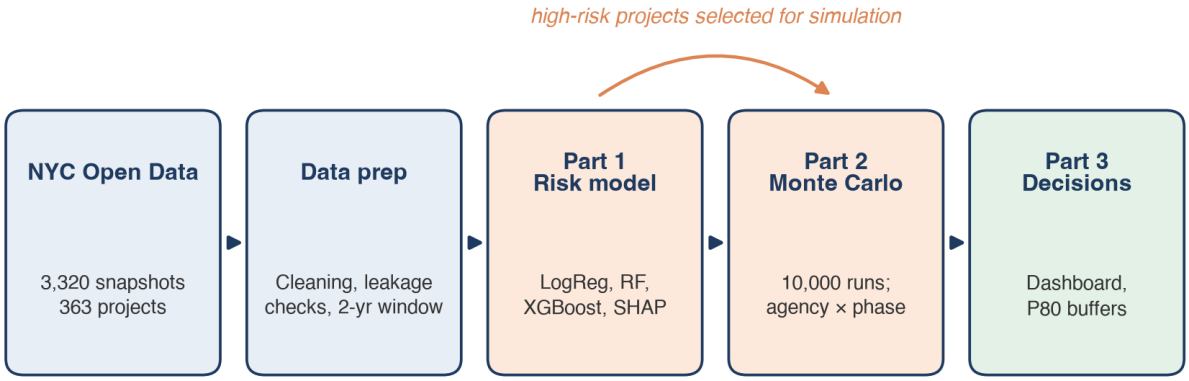


Figure 2. End-to-end pipeline. Part 1 scores every project; the highest-risk active projects feed the Part 2 simulation; Part 3 turns both into dashboards and recommendations.

5.1 Prediction points and targets (panel formulation)

To remove exposure bias, every outcome is measured over the same **fixed two-year horizon**. Rather than predicting only at a project's first report (which yields just 138 labelled projects), **every report *t* of every project is a prediction point**, giving 998 labelled examples from 183 projects. Delay and cost overrun are modelled as **separate targets**, since they behave differently (60% vs. 23% positive). With F the forecast completion and B the budget forecast:

$$y^{\text{delay}}_{i,t} = \mathbb{1}\left[F_{i,t+2y} - F_{i,t} > 365 \text{ days}\right], \qquad y^{\text{cost}}_{i,t} = \mathbb{1}\left[B_{i,t+2y}/B_{i,t} - 1 > 0.25\right] \tag{1}$$

The thresholds (one year, 25%) define materially troubled projects; a looser 90-day / 10% rule would label over 80% of projects positive. A combined flag (either event) is retained for comparison with the first-report baseline model.

5.2 Features and leakage control

Each prediction point uses only information available at report t. **Static features:** managing agency, category and current phase (one-hot encoded, rare groups merged), $\log_{10}$ budget, design start year, project age, planned total and remaining duration, and report year. **Track-record features:** slip and budget growth so far and over the last year, slip rate per day tracked, share of reports with a slip, years tracked, years in the current phase and number of phase changes. All lifetime totals and later snapshots are excluded; extreme ratios are capped and missing values median-imputed with indicators.

5.3 Models and evaluation

- **Models:** a majority-class baseline; L2-regularised logistic regression with class weighting (interpretable reference); Random Forest; and gradient-boosted trees (XGBoost; Chen and Guestrin, 2016) with shallow trees to limit overfitting.
- **Grouped validation:** because a project contributes several correlated rows, all of its rows are kept in the same fold (stratified group 5-fold CV, repeated 5 times). Without this, the model would partly "recognise" test projects and accuracy would be overstated.
- **Temporal holdout:** train on prediction points up to January 2017 and test on 2019–2021, with no overlap in time between training outcomes and test periods. This is the most deployment-realistic estimate.
- **Metrics:** ROC-AUC (primary), PR-AUC (important for the rarer overrun target), precision and recall; feature-set ablation (static vs. static + track record) shows where any gain comes from.
- **Interpretation:** odds ratios for logistic regression and SHAP values (Lundberg and Lee, 2017) for the tree model; agreement between the two strengthens conclusions about risk drivers.

5.4 Monte Carlo schedule and cost simulation

For each pair of consecutive reports, the movement of the forecast completion date F and of the budget B is attributed to the phase φ the project was in. Aggregating per project and phase gives an empirical **slip rate** r (days of slip per calendar day) and **cost-growth rate** c (budget growth per year):

$$r_{i,\phi} = \frac{\sum_{t \in \phi} (F_{i,t} - F_{i,t-1})}{\sum_{t \in \phi} \Delta t}, \qquad c_{i,\phi} = \frac{\sum_{t \in \phi} \Delta B_{i,t} / B_{i,t-1}}{\sum_{t \in \phi} \Delta t \;/\; 365.25} \tag{2}$$

A project's remaining time (last report t_0 to forecast completion) is split across its remaining phases in proportion to typical phase lengths measured from observed transitions (design ≈ 3.3 years, procurement ≈ 1.1, construction ≈ 2.8). In each of 10,000 runs, every remaining phase draws a real record k from the same agency and phase (pooled across agencies when fewer than 15 records exist), with a smoothed bootstrap (Silverman bandwidth h) to avoid artefacts from small pools. Simulated phase durations, completion date and cost growth are then:

$$\tilde{D}_\phi = D_\phi (1 + r_{k,\phi} + \varepsilon), \qquad \varepsilon \sim \mathcal{N}(0, h_\phi^2) \tag{3}$$

$$\tilde{T} = t_0 + \sum_\phi \tilde{D}_\phi, \qquad \tilde{G} = \sum_\phi c_{k,\phi}\, \tilde{D}_\phi \tag{4}$$

Outputs are the probability of meeting the reported date, P50 and P80 completion dates, schedule contingency (P80 minus plan) and budget contingency (P80 of G times the current budget). A one-at-a-time tornado analysis (each phase moved from its P10 to its P90 slip) identifies the dominant risk driver, and a fully correlated variant (one percentile shared across phases) bounds the effect of the independence assumption. Three high-risk active projects are simulated, one in each of design, procurement and construction, alongside a portfolio-level buffer table for each agency and starting phase.

5.5 Validating the simulator: historical back-test

The simulator is tested the way a weather forecast is: against what actually happened. From four past origins (January 2019 to March 2021), slip rates and phase lengths are re-estimated using **only data available at that origin**, and the simulator predicts how far each active project's forecast completion will move over the next two years. Calibration is measured by the share of real outcomes falling below each simulated percentile (ideally 80% below the P80), alongside the median error of the P50. Three variants are compared (independent phases, correlated phases, and blending in a project's own slip history), and a **phase-level quantile recalibration** is learned on the 2019 origins and evaluated on the unseen 2020–2021 origins.

5.6 Decision layer

Results are published as a three-page Power BI/Tableau dashboard (portfolio risk with a borough map, agency comparison, and simulation results with tornado charts), and synthesised into 3–5 recommendations, each traceable to a specific quantitative result.

6. Preliminary Results

A first full pass through the pipeline has been completed to establish feasibility. These results are preliminary and will be refined, stress-tested and reported in full in the final report.

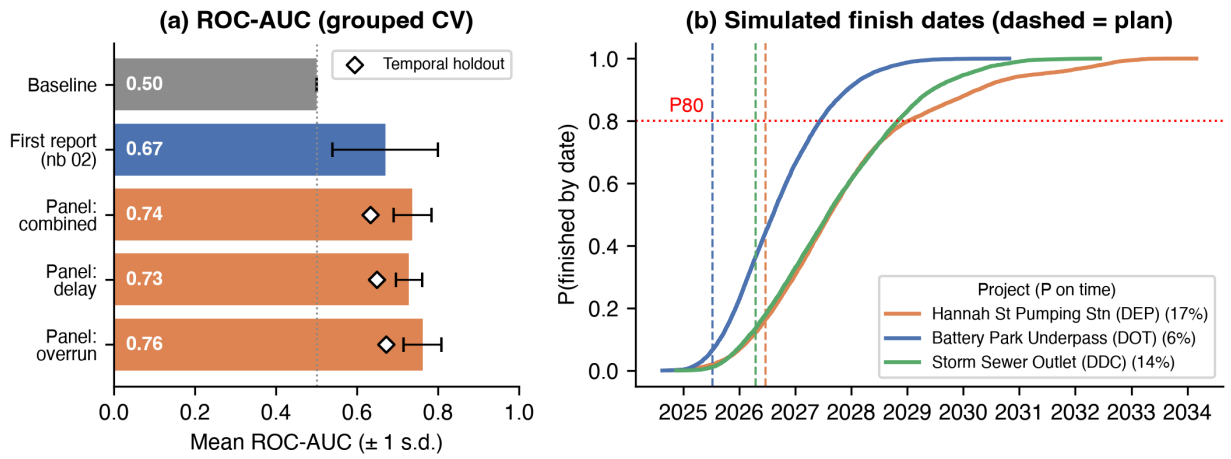


Figure 3. (a) ROC-AUC of the first-report model (138 projects, repeated stratified CV) and of the panel models (998 prediction points, grouped CV), mean ± 1 s.d.; diamonds show the stricter temporal holdout. (b) Simulated finish-date distributions for three high-risk projects; dashed lines mark each project's reported completion date, which falls deep in the left tail.

Table 3. Preliminary Monte Carlo results (10,000 runs per project; status as of January 2023).

Project (agency, phase)	P(on time)	P50 finish	P80 finish	Schedule contingency	Budget contingency (P80)
Battery Park Underpass (DOT, design)	6%	Aug 2026	Jun 2027	+703 days (+76%)	+49% (≈ $32M)
Storm Sewer Outlet, 224th St (DDC, procurement)	14%	Aug 2027	Nov 2028	+926 days (+77%)	+73% (≈ $114M)
Hannah St Pumping Station (DEP, construction)	17%	Aug 2027	Jan 2029	+928 days (+73%)	+25% (≈ $27M)

- **RQ1/H1 (supported):** projects first reported in design have 2.6× the odds of becoming high-risk. The strongest single warning sign is **little remaining planned time**: 97% of prediction points with under a year left on the plan slipped by more than a year. Large budget overruns are concentrated in design (31% of prediction points vs. 7% in construction). Contrary to intuition, **past slipping does not predict future slipping**, which is consistent with schedules being re-baselined after large slips.
- **RQ2/H2 (supported):** the panel models reach grouped-CV ROC-AUC **0.73 ± 0.03 for delay** (XGBoost) and **0.76 ± 0.05 for overrun** (logistic regression, PR-AUC 0.48 vs. a 0.23 base rate), against 0.67 ± 0.13 for the first-report model. Scored at a project's first report the panel models perform similarly to the baseline (≈ 0.67), so the gain comes from re-scoring as projects progress; the temporal holdout gives a more conservative 0.63–0.67.
- **RQ3/H3 (supported):** the simulated projects have only a 6–17% chance of meeting their reported dates. Portfolio simulations suggest adding roughly +80–90% to the remaining schedule for projects starting design and +65–75% for projects starting construction to reach 80% confidence.

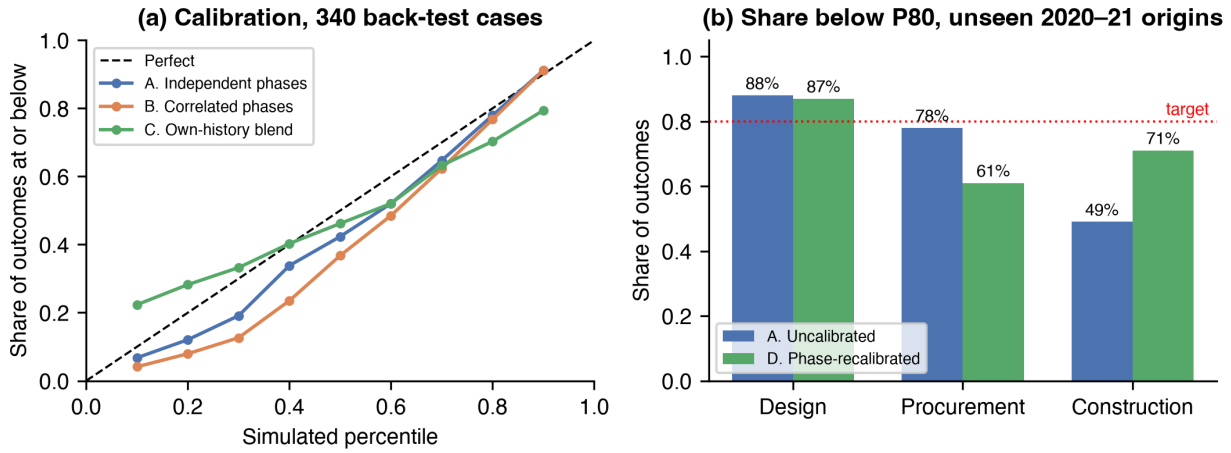


Figure 4. (a) Back-test calibration over 340 historical cases: the share of real two-year outcomes at or below each simulated percentile (the diagonal is perfect). The independent-phase simulator (A) tracks the diagonal closely at the upper percentiles used for contingency. (b) On unseen 2020–2021 origins, phase-level recalibration corrects the simulator's over-optimism for construction-phase projects.

Simulator validation. In the back-test, 78% of real outcomes fell at or below the simulated P80 (target 80%) and the P10–P90 range covered 84%. The simulator's median error was **266 days**, against **460 days** for the implicit forecast that the reported date is correct (42% lower). Correlated phases and own-history blending did not improve calibration; the latter made the simulator overconfident (57% P10–P90 coverage). The overall result hides a phase bias: the simulator under-predicts slip for construction-phase projects and over-predicts it for procurement. Recalibration learned on 2019 and tested on 2020–2021 reduced the median error from 284 to **210 days (−26%)** and raised the construction P80 hit rate from 49% to 71%.

7. Work Plan and Deliverables

Table 4. Work plan and current status.

Stage	Activities	Deliverable	Status
1. Data preparation	Cleaning, data audit, fixed-horizon target, phase-rate extraction	Notebook 01; clean tables	Complete
2. Exploratory analysis	Distributions, agency/category/phase comparisons, exposure analysis	EDA figures	Complete
3. Predictive modeling	First-report baseline; panel model with separate delay/overrun targets; grouped CV, temporal holdout, SHAP; risk scoring	Notebooks 02 and 05; risk lists	Complete; refining
4. Monte Carlo simulation	Project simulations, tornado, correlation sensitivity, portfolio buffers; historical back-test and recalibration	Notebooks 03 and 06; simulation outputs	Complete; refining
5. Dashboard	Three-page Power BI/Tableau build from prepared tables	Interactive dashboard	Planned
6. Reporting	Final report, recommendations, presentation	Report and slides	Planned

Planned refinements before the final report: hyperparameter tuning with nested grouped cross-validation, probability calibration of the risk scores, a regression formulation (predicting delay in days), applying the back-test recalibration to the project-level simulations, and, if time permits, extending the data with the newer Capital Projects Dashboard datasets.

8. Risks, Limitations and Ethical Considerations

Table 5. Key risks and mitigations.

Risk / limitation	Impact	Mitigation
Small sample (183 projects); correlated rows per project	High variance; overstated accuracy	Panel formulation (998 points); grouped CV; temporal holdout; regularisation; report uncertainty
Concept drift (e.g. COVID-era conditions)	Lower accuracy on future periods	Temporal holdout quantifies it (0.63–0.67); retrain as new reports arrive
Data ends January 2023; few completed projects	Outcomes are "to date", not final	Fixed-horizon target; frame results as as-of-2023; state clearly
Leakage and exposure bias in raw fields	Inflated, misleading accuracy	Point-in-time features only; fixed window; audit documented
Agency reporting gaps (SCA, FDNY); missing borough	Biased estimates for those groups	Exclude from delay model; label clearly in dashboard
Simulation assumptions (phase independence, no close-out, stationarity)	Mis-stated tail risk	Back-tested on 340 cases; phase recalibration; correlated-phase sensitivity bound

Ethics. The data are public and contain no personal information. Risk scores describe statistical patterns, not the performance of individual staff or contractors, and are intended to prioritise review, not to penalise agencies. Model limitations and uncertainty will be reported alongside every result.

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